Reply to the Referees

Manuscript QT10218 — "Non-Local Magic Resources for Fermionic Gaussian States"
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We thank both Referees for their careful reading of the revised manuscript. Below we reply point by point and list all changes of content. In this version, the manuscript is formatted as a Letter: the main results are in the main text, the technical details in the End Matter, and the remaining Appendices in the Supplemental Material (SM), whose content is otherwise unchanged except for the modifications listed below; Sec. S5 collects the numerical details on SYK_2 and random circuits previously given in the main text. The table at the end maps the equation numbers quoted in the reports to the new ones.

Referee A

> *I believe that the current statement of theorem 1 is explicitly false. [...] the $T_{\pi_{AB}}$ in equations (5) and (7) is a non-Gaussian permutation of qubits while in the appendix (e.g. from equation (50)) we have only Gaussian operations. [...] Another change which would fix the theorem is to restrict to only considering contiguous partitions (so A consists of the first l modes and B the rest).*

Reply. We thank the Referee for this important observation and for the explicit counterexample, which we have verified. We agree, and, as suggested, we now consider only the bipartition into the contiguous blocks $A = \{1, \dots, \ell\}$ and $B = \{\ell+1, \dots, N\}$ of the Jordan–Wigner chain, both in Theorem 1 and, for consistency, in the definitions of the non-local and FNL magic [Eqs. (4)–(5), now Eqs. (2)–(3)], so that no qubit permutation $T_{\pi_{AB}}$ appears. In this case $U_{O_A} \otimes U_{O_B}$ coincides, up to a phase, with the Gaussian unitary of the whole chain associated with $O_A \oplus \det(O_A)\, O_B$; hence the minimization in Eq. (5) [now Eq. (3)] is exactly the maximization over the orbit of Eq. (50) [now Eq. (S5)] solved in the proof. This argument is now given in the End Matter ("Local Gaussian orbit") and below Eq. (S5) in the SM, together with a remark that, as shown by the Referee's example, the identification fails for other bipartitions. The definitions, the statement of Theorem 1 and the abstract have been modified accordingly. The restriction does not affect any of the applications: for Haar-random states and SYK_2 eigenstates the statistics of Γ_A is invariant under mode permutations, while for the XY chain, the quenches, the random circuits and the variational checks we used contiguous blocks $A = \{1, \dots, \ell\}$ [authors: please confirm for the random-circuit data of Fig. 3(d)]. With this restriction, Eq. (4) still provides the upper bound $M_2^{\mathrm{NL}} \le M_2^{\mathrm{FNL}} = \sum_i \mathfrak{m}_2(\lambda_i^2)$ on the unrestricted non-local magic.

> *[...] although the manuscript lacks any AI disclosure statement, in my opinion the new appendix appears to have significant LLM input. [...] overall it would be nice to have some intuition about why we are making the moves we are making [...].*

Reply. [Authors: adapt this paragraph to your actual use of AI tools.] We have added a statement on the use of AI tools in the Acknowledgments, and we have revised all the passages indicated by the Referee (points 2–14 below). Moreover, at the beginning of Sec. S1 we added a paragraph explaining the logic of the proof, i.e., why each step is taken, which complements the "Odyssey" figure; a shorter version is in the End Matter.

2. *I think the middle line of equation (124) isn't exactly right [...].*

Reply. Corrected: the expectation value now acts on the whole product, and the independence of the rows is used only in the last step [Eq. (S75)].

3. *The text under equation (60) states that the equation is just Cauchy-Binet but in fact it requires some extra reasoning [...].*

Reply. We now state that the identity follows from the principal-minor expansion $\det(\mathbb{I}+Y) = \sum_J \det Y_{J,J}$ applied to $Y = X^\dagger X$, combined with the Cauchy–Binet formula for each principal minor of $X^\dagger X$ [below Eq. (S16)].

4. *Lemma 1 is proved with the line "The determinantal generating function gives" [...] it would be nice to show the proof of this if so.*

Reply. We now derive Eq. (59) [now Eq. (S15)] explicitly: writing $(-1)^{|S\cap I|} = \prod_{i\in I}(1-2\,\mathbf{1}_{i\in S})$, expanding the product and using that the inclusion probabilities of the determinantal process are $\sum_{S\supseteq T} p_S = \det K_T$ (with a reference), the principal-minor expansion gives $\det(\mathbb{I} - 2K_I)$. The expression "generating function" has been removed.

5. *I would be interested in an explanation of what the phrase "The transpose, rather than the adjoint, is what remembers the particle–hole reality constraint in the fermionic application" actually means [...].*

Reply. The sentence meant that the relevant object is the transpose overlap $\Sigma = Q^\mathsf{T} Q$ (the adjoint overlap $Q^\dagger Q = \mathbb{I}$ being trivial), because in the application (Proposition S2) the reality of the covariance matrix translates into the complex orthogonality $Z^\mathsf{T} Z = \mathbb{I}$ of the BCS matrix, which makes Σ explicitly computable. We replaced the sentence with this explanation.

6. *In the phrase "The two chains coincide term by term, proving Eq. (108)", I am not sure what the chains are.*

Reply. The two "chains" were the two sequences of equalities for the two sides of Eq. (108). This part has been replaced by the Referee's simpler proof (point 11).

7. *The proof between (80) and (81) is unnecessarily convoluted. [...]*

Reply. Simplified as suggested: $0 < \mathcal{R} \leq \mathbb{I}$ gives $\|\mathcal{R}\| \leq 1$ and $|e_1^\dagger \mathcal{R} u| \leq \|u\| \leq 1$ [after Eq. (S36)].

8. *[...] In my opinion equation (147) is already sufficient, (148) is unnecessary and somewhat confusing.*

Reply. Eq. (148) has been removed.

9. *The text "In this subsection we rewrite it exactly — no inequality is involved — [...]" is very LLM-ish and slightly confusing [...].*

Reply. Rephrased (beginning of Sec. S1.5).

10. *Just above equation (136), the word invariant only has one v in it.*

Reply. Corrected; we also clarified the sentence, which refers to the invariance of the probabilities under a unitary change of column gauge.

11. *The proof of lemma 5 and especially Equation (115) is (subjectively, for me) borderline unreadable. [...] I sketched this proof in section 4.*

Reply. We thank the Referee for the elegant alternative proof, which we have checked and adopted in Lemma S5 [Eqs. (S64)–(S67)], acknowledging the Referees in the Acknowledgments. (In Eqs. (20)–(24) of the report, $\det(\mathbb{I} - H_S H_S^\mathsf{T})$ should read $\det(\mathbb{I} + H_S H_S^\mathsf{T})$ and the sums run over $A \subseteq S$, $B \subseteq S^c$; the conclusion is unchanged.) The sign-average identity previously established in the proof of Lemma 5 is now derived where it is used, in Proposition S3.

12. *I don't think the text "with no factorial normalization." in theorem 2 is necessary [...].*

Reply. Removed.

13. *[...] the notations $\|\cdot\|_{\mathrm{HS}}$ and $\|\cdot\|_2$ are used for exactly the same thing.*

Reply. We now use $\|\cdot\|_{\mathrm{HS}}$ throughout Sec. S1.

14. *The notation from equation (145) is technically consistent, but unfortunate [...].*

Reply. We now denote the creation operator by $c_x^\dagger = \sum_j x_j c_j^\dagger$, and its adjoint by c_x, throughout the proof of Theorem S1 [Eqs. (S96)–(S104)].

We also corrected a few further typos in the Appendix (e.g., in the definition of the coefficient $[z_1 \cdots z_n]f$ and in the Remark after Theorem S1). Finally, independently of the reports, we corrected the early-time expansion for random Gaussian circuits (which must be performed around $\lambda = 1$, the circuits starting from a product state), and we specified the time units of the quench formulas (those of Refs. [Calabrese–Cardy 2005, Fagotti–Calabrese 2008], i.e., evolution generated by $H(\mu)/2$) [authors: check against the code used for Fig. 3]; neither change affects the results.

Referee B

Equation (11) of the revised manuscript appears inconsistent with the previous result derived by Bianchi, Hackl, and Kieburg [...] Eq. (11) of the manuscript therefore contains an additional factor $\prod_m (1 - x_m)^{-1/2}$.

Reply. We thank the Referee for the positive assessment of Theorem 1 and for spotting this inconsistency. (Following Referee A, Theorem 1 is now stated for bipartitions into two contiguous blocks; see our reply to Referee A.) The additional factor is a typo: the correct weight is the one of Bianchi, Hackl and Kieburg, $x^{-1/2}(1-x)^{N-2\ell}$, which we now cite. Point by point:

1. It is a typographical error, which also propagated to the parameters of the Jacobi polynomials and to h_n in Eqs. (14)–(15); it does not affect the $N \to \infty$ results (point 4) [authors: confirm that no figure used the finite-N formula with the old weight].

2. We corrected Eqs. (14)–(15) consistently: $p_n(x) = \mathrm{P}_n^{(-1/2, N-2\ell)}(1-2x)/\sqrt{h_n}$, with $h_n = \Gamma(n+\frac{1}{2})\Gamma(n+N-2\ell+1)/[(2n+N-2\ell+\frac{1}{2})\Gamma(n+1)\Gamma(n+N-2\ell+\frac{1}{2})]$ [now Eqs. (13)–(14) in the End Matter; the corrected weight is Eq. (12)].

3. The analytical curve in Fig. 1(a) is the $N \to \infty$ result, Eq. (18) [now Eq. (6)], as stated in the caption; the data are obtained from Haar-random $O \in \mathrm{O}(2N)$, $\Gamma = O\Gamma_0 O^\mathsf{T}$ [authors: confirm sampling method]. The sentence of the previous version stating that Eq. (14) [now Eq. (13)] was benchmarked in Fig. 1(a) was inaccurate, since Fig. 1(a) only shows the $N \to \infty$ curve; it has been replaced (End Matter, below Eq. (14)). As suggested, we checked the corrected finite-N formula against direct sampling (for $\ell = 1, 2$ and several $N \geq 2\ell$), finding agreement within statistical errors, whereas the weight of the previous Eq. (11) is clearly incompatible with the sampled data.

4. The thermodynamic-limit results, Eqs. (16)–(18), including the half-partition value $2 - \log_2(2+\sqrt{3})$, are unchanged, since the exponent of $(1-x)$ only affects the $O(1/N)$ corrections to $\overline{M}_2^{\mathrm{FNL}}/N$.

Equation numbers

previous	this version	previous	this version
Eq. (5)	Eq. (3)	Eq. (57)	Eq. (S13)
Eqs. (6), (7)	Eq. (4), Theorem 1	Eqs. (59), (60)	Eqs. (S15), (S16)
Eq. (11)	Eq. (12)	Eqs. (80), (81)	Eq. (S36) and text below
Eqs. (14), (15)	Eqs. (13), (14)	Eq. (108), Lemma 5	Eq. (S63), Lemma S5
Eqs. (16)–(18)	text above Eq. (6), Eq. (6)	Eqs. (115), (116)	replaced by Eqs. (S64)–(S67)
Eq. (50)	Eq. (S5)	Eq. (124)	Eq. (S75)
Lemma 1	Lemma S1	Eq. (136)	Eq. (S87)
Theorems 2, 3	Theorems S1, S2	Eqs. (145), (147), (148)	Eqs. (S96), (S98), removed